MIL-D-46206D (MU)
10 August 1966
SUPERSEDING
MIL-D-46206C (MU)
18 September 1964

MILITARY SPECIFICATION

DELAY OR NON-DELAY ELEMENT, M9 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the quality assurance provisions, and requirements not covered by the drawing, for the loading, assembling and packing for one type of delay or non-delay element designated as Delay or Non-delay Element, M9 (T2E3) (see 6.5).

1.2 Classification.-The M9 elements shall be of the following types as specified below (see 6.1).

Type I - Non-Delay
Type II - .010 Second Delay
Type III - .025 Second Delay
Type IV - .050 Second Delay
Type V - .100 Second Delay
Type VI - .250 Second Delay

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-A-2550 - Ammunition and Special Weapons; General Specification for.
MIL-P-20444 - Primer, Percussion, M42, Loading, Assembling, and Packing.

FSC 1375

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MIL-I-45607 - Inspection, Supply and Maintenance of.
MIL-D-45608 - Designs and Drawings, Ordnance Inspection Equipment
MIL-C-45662 - Calibration of Standards.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-1167 - Ammunition Data Cards.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1234 - Pyrotechnic Sampling, Inspection and Testing.
MIL-STD-1235 - Single and Multilevel Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

U.S. ARMY MUNITION COMMAND

8796666 - Box, Packing, Ammunition for Element, Delay, or Non-Delay.
8796667 - Carton, Packing, Ammunition, For Element, Delay, or Non-delay, M9.
8838187 - Delay Assembly
8838195 - Delay Cup Assembly
8838197 - Element, Delay or Non-Delay, M9 Assembly
8885299 - Non-Delay Assembly

PUBLICATIONS

ORDNANCE CORPS

IEL-8838197 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with the applicable drawings and specifications.

3.2 Elements.-The delay or non-delay element assemblies shall comply with all the requirements specified on Drawing (dwg.) 8838197, (as applicable) and with all requirements specified in applicable specifications.

3.3 Moisture content.-Moisture content of igniter composition or of delay charge composition at the loading station at the time of loading shall be not greater than 0.04 percent when tested as specified in 4.3.1.

3.4 Water soluble chloride content.-The water soluble chloride content of the igniter composition or of the delay charge composition shall be not greater than 0.05 percent when determined as specified in 4.3.2.

3.5 Static functioning (at ambient temperature).

3.5.1 Non-delay.-The functioning time of the non-delay element shall be 0.001 second (sec.), maximum, when tested as specified in 4.3.5.

3.5.2 Delay.-The delay elements shall function within the tolerance limits specified in Table I when tested as specified 4.3.5.

TABLE I

(all times are in seconds)

DELAY	LOWER LIMIT	UPPER LIMIT
.010	.0025	.0175
.025	.0145	.0355
.050	.0323	.0678
.100	.0609	.1391
.250	.1850	.3250

3.6 Impact functioning.-The delay assemblies, after temperature conditioning at 160 degrees F. or minus 65 degrees F., shall not fail to function within the applicable delay limits of 3.5 when subjected to an impact force of 7000 g's. Testing shall be as specified in 4.3.6.

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3.7 Workmanship.-Loading procedures shall be in conformance with the best industrial practices. All components and subassemblies shall be free of chips, dirt, grease, rust, corrosion products, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind

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his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor. Each lot shall contain:

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a. Primers from not more than two lots from one manufacturer bearing the same interfix lot number.
b. Relay elements from not more than two lots from one manufacturer bearing the same interfix lot number.
c. Igniter composition from one lot from one manufacturer.
d. Delay charge composition from one lot from one manufacturer.
e. Metal parts of one lot interfix number from one manufacturer.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used (group or individual) that method will be used for all of the CD's listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.2.2.1 Assembly, Non-delay (see dwg. 8885299 covering a detail of dwg. 8838197).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Relay element missing or inverted	Visual	01001
102.	Primer missing or inverted	Visual	01002
Minor:	AQL 1.00 percent		
201.	Either crimp not full 360 degrees	Visual	01003
202.	Length, maximum (max.)	Gage	01004

4.2.2.2 Assembly, Delay Cup (see dwg. 8838195 covering a detail of dwg. 8838197).

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Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cup split or cracked	Visual	02001
Major:	AQL 0.40 percent		
101.	Relay element inverted	Visual	02002
102.	Charge missing	Visual	02003
Minor:	AQL 1.00 percent		
201.	Varnish missing from relay	Visual	02004
202.	Evidence of poor workmanship (see 3.7)	Visual	02005

4.2.2.3 Assembly, Delay, prior to coating threads (see dwg. 8838187 covering a detail of dwg. 8838197).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Primer missing or inverted in primer holder assembly	Visual	03001
102.	Baffle missing	Visual	03002
103.	Flash hole obstructed	Visual	03003
Minor:	AQL 1.5 percent		
201.	Crimp on primer end not full 360 degrees	Visual	03004
202.	Coating missing on primer	Visual	03005
203.	Evidence of poor workmanship (see 3.7)	Visual	03006

4.2.2.4 Assembly, Delay (see dwg. 8838187 covering a detail of dwg. 8838197).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			

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Major:	AQL 0.40 percent		
101.	Total length, max.....	Gage	04001
Minor:	AQL 1.5 percent		
201.	Assembly loose.....	Manual	04002
202.	Sealing compound missing from threads.....	Visual	04003
203.	Evidence of poor workmanship (see 3.7).....	Visual	04004

4.2.2.5 Assembly, Element, Delay and Non-Delay (see dwg. 8838197).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Total length, max.....	Gage	05001
102.	Diameter, max.....	Gage	05002
103.	Delay or non-delay assembly missing.....	Visual	05003
Minor:	AQL 2.5 percent		
201.	Washer above flush.....	Visual	05004
202.	Stake of washer missing.....	Visual	05005
203.	Stake of washer above flush....	Visual	05006
204.	Marking incorrect, misleading or unidentifiable.....	Visual	05007
205.	Evidence of poor workmanship (see 3.7).....	Visual	05008

4.2.2.6 Carton, prior to sealing (see dwg. 8796667).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Packing component missing.....	Visual	06001

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Minor:	AQL 1.0 percent		
201.	Separator damaged to extent that elements touch each other..	Visual	06002
202.	Element missing.....	Visual	06003

4.2.2.7 Carton, sealed (see dwg. 8796667).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.5 percent		
201.	Sealing strip torn, badly wrinkled or incomplete.....	Visual	07001
202.	Glued outer carton flaps not securely sealed to inner flaps.	Visual	07002
203.	Carton torn, cut or punctured..	Visual	07003
204.	Marking, incorrect, misleading or unidentifiable.....	Visual	07004
205.	Contents loose.....	Manual	07005

4.2.2.8 Sealed Barrier Bag (see dwg. 8796667).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Bagged item fails heat seal test.	Test (see 4.3.7)	08001
102.	Bag punctured, torn or cut.....	Visual	08002
103.	No area for subsequent seals (a minimum of two subsequent seals).	Visual	08003
Minor:	AQL 0.65 percent		
201.	Marking incorrect, misleading or unidentifiable.....	Visual	08004

4.2.2.9 Sealed wood packing box (see dwg. 8796666).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined.

Major:	AQL 0.65 percent		
101.	Box broken or split exposing contents	Visual	09001
102.	Hardware broken or loose	Visual-Manual	09002
103.	Strapping broken or loose	Visual-Manual	09003
Minor:	AQL 2.5 percent		
201.	Hardware improperly engaged	Visual	09004
202.	Strapping improperly assembled.	Visual	09005
203.	Car seal unsealed or improperly positioned	Visual	09006
204.	Marking incorrect, misleading or unidentifiable	Visual	09007
205.	Contents loose when tilted	Manual	09008

4.2.3 Testing

4.2.3.1 Sample for delay or igniter charge composition analysis.- A sample of fifty (50) grams shall be selected from each lot or batch of delay or igniter charge composition for the performance of the following analysis and determinations. Specimens for the following tests shall be selected from this sample, which shall be stored in an air tight container until all of the tests have been completed. This sample will provide additional specimens for performing tests which have been improperly performed or terminated due to circumstances beyond the control of the laboratory.

4.2.3.1.1 Composition of delay charge (see dwg. 8838195) - Major defect - Code No. 02005 or igniter charge (see dwg. 8838195) Major defect - Code No. 02006.-A portion of four (4) grams of the sample shall be selected for either analysis which shall be performed as specified in 4.3.3 or 4.3.4, as applicable. If any portion fails to meet one or more of the applicable requirements, the represented lot or batch shall be rejected.

4.2.3.1.2 Determination of moisture content (see 3.3) - Delay charge - Major defect - Code No. 02007 or igniter charge - Major defect - Code No. 02008.-A portion of twenty (20) grams of the sample shall be selected for this determination which shall be performed as specified in 4.3.1. If any portion fails to meet the requirement, the represented lot or batch shall be rejected.

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4.2.3.1.3 Determination of water soluble chloride content (see 3.4) - Delay charge - Major defect - Code No. 02009 or igniter charge - Major defect - Code No. 02010.-The portion for this determination shall be of the same weight as that of 4.2.3.1.2. The determination shall be performed as specified in 4.3.2. If any portion fails to meet the requirement, the represented lot or batch shall be rejected.

4.2.3.2 Static functioning

4.2.3.2.1 Non-Delay (see 3.5.1) - Major defect - Code No. 05009.-Duds - Major defects - Code No. 05010.-A sample of fifty (50) non-delay elements shall be selected from each lot and tested as specified in 4.3.5 using test equipment in accordance with 4.2.4. The lot shall be rejected if more than one (1) assembly fails to function (dud), or if more than two (2) samples exceed the time specified. If two samples exceed the specified time (excluding duds) a second sample of 50 non-delay elements may be selected and tested as specified in 4.3.5. Testing of the second sample for Government acceptance, shall be stopped and the lot shall be rejected at any time during the second test if two elements fail to function within the required time or at any time when a combined number of 2 or more duds are observed in the first and second samples.

4.2.3.2.2 Delay (see 3.5.2) - Functioning time - Major defect - Code No. 05011 - Duds - Major defect - Code No. 05010.-A sample of fifty (50) delay elements shall be selected from each lot and shall be tested as specified in 4.3.5 using test equipment specified in 4.2.4. The average functioning time and the sample standard deviation shall be calculated for each lot tested (see 6.4). The lot shall be rejected if any sample fails to meet the following:

a. The average functioning time plus the product of 2.08 and the sample standard deviation exceeds the upper limit specified, or

b. The average functioning time minus the product of 2.08 and the sample standard deviation is less than the lower limit specified.

4.2.3.2.3 Reduced testing (static functioning).-After three consecutive lots have met the criteria of 4.2.3.2.2, static functioning testing shall be performed on every fifth lot thereafter.

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4.2.3.3 Impact functioning, delay (see 3.6).-Functioning at 160 degrees Fahrenheit (F.).-Major defect - Code No. 05012; Functioning at minus 65 degrees F. - Code No. 05013; Duds - major defect - Code No. 05010.-A sample of sixty (60) delay elements shall be selected from each lot and divided into two equal groups. One group shall be conditioned at 160 degrees F., and the other group at minus 65 degrees F. The assemblies shall be tested as specified in 4.3.6 using test equipment specified in 4.2.4. The average functioning time shall be calculated for each temperature group (see 6.4). The lot shall be rejected if any or both temperature groups fail to meet the following:

a. The average functioning time plus the product of 2.00 and the sample standard deviation exceed the upper limit specified, or

b. The average functioning time minus the product of 2.00 and the sample standard deviation is less than the lower limit specified, or.

c. If two (2) or more duds are encountered during the entire test sequence in 4.2.3.2.2 and 4.2.3.3.

4.2.3.4 Check test for primer deterioration.-If the total time between original acceptance of any primer lot and the assembly of the lot into the element exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since the last previous test, the primers shall be subjected to and must satisfactorily pass the check test specified in the primer loading specification MIL-P-20444 before the primers are assembled into the Delay or Non-Delay Element (see 6.6).

4.2.4 Inspection Equipment.-Index of Inspection Equipment Number IEL-8838197 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction,survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and Maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

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4.2.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.2.4.4 Calibration.-The contractor shall have available a set of master gages, standards and appropriate instruments for regularly scheduled calibrations of his acceptance inspection equipment. Records of such regularly scheduled calibrations shall be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments shall be in accordance with Specification MIL-C-45622.

4.3 Test methods and procedures

4.3.1 Determination of moisture content of delay charge composition; or of igniter composition.-An accurately weighed portion of 10 grams shall be taken from the sample of 4.2.3.1.2 for this test. Test shall be conducted in accordance with Method 102.1 of Standard MIL-STD-1234. Determination shall be made for conformance with 3.3.

4.3.2 Determination of water soluble chloride content of delay charge composition or of igniter composition.-Dissolve approximately 1 gram accurately weighed from the sample in about 60 millitiers of hot distilled water. Cool the solution to room temperature, transfer to 100 ml volumetric flask and dilute to volume. Pipette 10 ml of this solution into a Nessler tube, and dilute to 30 ml with distilled water. Add 30 ml of distilled water to another Nessler tube. Add 1 ml of concentrated nitric acid and 1 ml of 0.1 normal silver nitrate solution to the sample and blank. To the blank, add a sufficient amount of sodium chloride solution (0.825 gram per liter) that will match the turbidity of the sample. Allow the solutions to stand 15 minutes in the dark, then, visually compare the sample with the standard. After the turbidities are matched, the percent chlorides is calculated and compared with the requirement.

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Calculation:

$$\text{Percent of Chlorides} = \frac{0.50\ V}{W}$$

where:
V = volume of standard solution required to match the turbidity of the sample,
W = weight of sample in aliquot, gm.

4.3.3 Determination of delay charge composition F33B and of igniter composition, Type I.

4.3.3.1 Silica (Diatomaceous earth).-Transfer an accurately-weighed sample of one gram from 4.2.3.1.1 to a 250 millilitre (ml.) Erlenmeyer flask, and add 20 ml. of 85 percent sulfuric acid (specific gravity 1.77). Heat gently at first to dissolve the ferric oxide and then more vigorously to dissolve the zirconium. Continue the heating until fumes of sulfuric acid are evolved. Ferric sulfate will separate from the solution. Cool and cautiously dilute with 50 ml. of water to dissolve the salts. If any red or gray particles remain undissolved, again heat to fumes of sulfuric acid. Cool, and dilute to 150 ml. with water. Filter through a No. 42 Whatman filter paper (or its equivalent), and wash thoroughly with about 40 ml. of warm sulfuric acid (1:100). Collect the filtrate in a 250 ml. volumetric flask. Place the filter paper in a platinum crucible, and dry in an oven at 100 to 110 degrees Centrigrade (C.) for 2 hours. Then place the crucible and its contents in a cold muffle furance, and raise the temperature to read 300 to 400 degrees C. until the paper has been completely charred. Raise the temperature to 1000 plus or minus 50 degrees C., and ignite for 1/2 hour. Cool in a desiccator, and weigh. To the ignited residue, add 5 ml. of hydrofluoric acid and one drop of concentrated sulfuric acid. Evaporate the acid, ignite and reweigh. The difference in weight, before and after adding the hydrofluoric acid, is due to the loss of silica and this difference should be inserted in the numerator of the following equation:

$$\text{Percentage of silica} = \frac{(\text{Loss in weight})\ 100}{(\text{Weight of samples})}$$

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Fuse the nonvolatile residue with one gram of potassium bisulfate. Dissolve the cooled melt in hot water, and add the solution to the 250 ml. flask that contains the silica filtrate. Dilute to the mark, mix, and reserve the determination of ferric oxide and zirconium.

4.3.3.2 Ferric oxide.-Take a 50 ml. aliquot from the 250 ml. flask containing the silica filtrate from 4.3.3.1 and add 50 ml. of concentrated hydrofluoric acid. Heat the solution nearly to boiling and add stannous chloride ($SnCl_2$) solution prepared by dissolving (3 grams stannous Chloride ($SnCl_2$) in 10 ml. of hydrochloric acid diluted to 20 ml. with water) dropwise until the color of the solution changes from yellow to light green. If this color change is not sharp, discard the solution, take a new aliquot, evaporate to 10-15 ml., add an equal volume of concentrated hydrochloric acid and proceed with the $SnCl_2$ reduction as described above. Add one or two drops of $SnCl_2$ in excess, cool to room temperature and rapidly add 10 ml. of mercuric chloride solution (saturated aqueous solution of mercuric chloride). If the precipitate formed is very heavy, or is other than white, too much $SnCl_2$ has been used and the reduction process must be repeated. Allow the solution to stand for about five minutes and then transfer to a 600 ml. beaker with the aid of about 250 ml. of water. Cautiously add 15 ml. of sulfuric acid and five ml. of phosphoric acid. Titrate slowly with 0.05 normal standard potassium dichromate solution until the pure green color changes to a gray-green. Then add the dichromate one drop at a time until the first tinge of purple or violet-blue appears.

$$\text{Percentage of Ferric Oxide, Ferric} = \frac{VN\ (39.92)}{W}$$

where:

V = ml. of standard potassium dichromate solution used.
N = normality of standard potassium dichromate solution (0.05)
W = weight of sample

4.3.3.3 Zirconium.-Transfer a 50 ml. aliquot of the silica filtrate from 4.3.3.1 to a 250 ml. beaker. Add 50 ml. of mandelic acid solution (16 grams in 100 ml. of water), and place on a steam bath for one hour. Remove from the steam bath, and let stand for at least 3 hours. Filter the solution through a 11 centimeter No. 40 Whatman filter paper (or its equivalent) and wash the precipitate with a solution containing 5 grams of mandelic acid and 2 ml. of hydrochloric acid per 100 ml. of water. Use a policeman to remove the precipitate adhering to the stirring rod and the beaker. Place the filter paper in a weighed platinum crucible, and dry in an oven at 100 to 110 degrees C. for 2 hours. Then

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place the crucible and its contents in a cold muffle furnace and raise the temperature to 300 to 400 degrees C. until the paper has been completely charred. Raise the temperature to 1000 plus or minus 50 degrees C., and ignite for 1/2 hour. Cool in a desiccator and weigh as zirconium oxide. However, if the residue is brown, add 1 gram of potassium sulfate, mix well, and fuze. Cool, dissolve in hot water, add 5 ml. of concentrated hydrochloric acid, dilute to 50 ml., and proceed as before by precipitating with mandelic acid.

$$\text{Percentage of zirconium} = \frac{(\text{Weight of zirconium oxide})\ 370.2}{\text{Weight of sample}}$$

4.3.3.4 Acetate resin.-The acetate resin content shall be calculated as the difference between 100 percent and the sum of the determinations of 4.3.3.1, 4.3.3.2 and 4.3.3.3.

4.3.4 Determination of delay charge composition Z-2A or Z-2B.-Determination of igniter composition Type II.

4.3.4.1 Barium chromate.-Transfer an accurately weighed sample of one gram from 4.2.3.1.1 to a 250 ml. beaker. Add 50 ml. of dilute perchloric acid (1:20), and heat on a steam bath until the barium chromate is dissolved. Dilute to 100 ml. with water, and filter through a No. 42 Whatman filter paper (or its equivalent). Collect the filtrate in a 250 ml. volumetric flask. Wash the residue with 25 ml. of dilute perchloric acid (1:100) and then 50 ml. of water. Reserve the filter paper and residue for the zirconium determination. Fill the volumetric flask to the mark, mix, and take a 50 ml. aliquot for the chromate determination. Dilute to 250 ml., and add 10 ml. of dilute sulfuric acid (1:4) and 10 ml. of phosphoric acid (1:1). Add a measured excess consisting of approximately 30 ml. of standardized ferrous ammonium sulfate solution (about 0.05 N) and 6 to 8 drops of sodium diphenylamine sulfonate indicator solution (0.2 gram per 100 ml. of water). Titrate the excess ferrous iron with 0.05 N standard potassium dichromate solution. Add the dichromate solution slowly with stirring until the pure green color changes to gray-green. Then add the dichromate one drop at a time until the first tinge of purple or violet-blue appears.

$$\text{Percentage of barium chromate} = \frac{(VN) \text{ minus } (CM)\ 42.23}{W}$$

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where:

V = ml. of standardized ferrous ammonium sulfate solution used.
C = ml. of standard potassium dichromate solution
N = normality of standardized ferrous ammonium sulfate solution
M = normality of standard potassium dichromate solution
W = weight of sample

4.3.4.2 Zirconium.-Transfer the filter paper with the zirconium from the barium chromate determination in 4.3.4.1 to a 250 ml. Erlenmeyer flask. Add 10 ml. of concentrated sulfuric acid and 5 ml. of concentrated nitric acid to dissolve the zirconium and destroy the filter paper. Heat to fumes of sulfuric acid, adding more nitric acid dropwise if the solution darkens. Cool and cautiously wash down the sides of the flask with about 25 ml. of water. If any insoluble material remains after warming, again heat to fumes. The solution of the zirconium may be speeded by heating the flask directly over a burner with rapid swirling until thick fumes of sulfuric acid are formed. A clear solution should remain after the salts dissolve. Dilute to 200 ml. in a volumetric flask. Transfer a 50 ml. aliquot to a 250 ml. beaker. Add 50 ml. of mandelic acid solution (16 grams per 100 ml. of water). Let stand on a steam bath for one hour. Remove from the steam bath, and set aside for at least 3 hours. Filter the solution through an 11-cm, No. 40 Whatman filter paper (or its equivalent), and wash the precipitate with a solution containing 5 grams of mandelic acid and 2 ml. of hydrochloric acid in 100 ml. of water. Use a policeman to remove the precipitate adhering to the stirring rod and the beaker. Place the filter paper in a weighed platinum crucible, and dry in an oven at 100 to 200 degrees C. for 2 hours. Then place the crucible and its contents in a cold muffle furnace and raise the temperature to 300 to 400 degrees C. until the paper has been completely charred. Raise the temperature to 1000 plus or minus 50 degrees C., and ignite for 1/2 hour. Cool in a desiccator, and weigh as zirconium oxide.

$$\text{Percentage of zirconium} = \frac{(\text{Weight of zirconium oxide})\ 296.1}{(\text{Weight of samples})}$$

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4.3.4.3 Acetate resin.-Determination of acetate resin shall be made by subtracting the sum of the determinations made in 4.3.4.1, and 4.3.4.2 from the total specified on applicable drawing for the igniter composition.

4.3.5 Static functioning (functioning time).-The element shall be conditioned at ambient temperature, 70 degrees plus or minus 3 degrees F. for a minimum of 6 hours. The assembly shall be assembled to the applicable test equipment and tested by dropping a steel ball weighing 3.95 plus or minus .02 ounces on the firing pin of the test fixture from a height of 8 plus or minus 1/8 inches. (The height of the drop shall be measured as the height from the top of the firing pin to the bottom of the ball). The time shall be measured and recorded or the result recorded as a dud if the assembly fails to function. Functioning time is defined as the elapsed time from when the ball impacts the firing pin until the assembly detonates.

4.3.6 Impact functioning testing.-The sample units shall be temperature conditioned at the required temperature plus or minus 3 degrees for 6 hours, minimum. To maintain temperature until tested, the units shall each be conditioned in an insulating container with the cover off the containers during conditioning. (A low density plastic foam is suggested for container material.) Before removal from the chamber, the cover shall be placed on the container and shall not be removed until the unit is to be assembled into the test equipment. The test shall be performed with the least possible delay. Delay times shall be recorded (see 6.9). Any unit failing to meet the applicable delay requirement shall be classed defective.

4.3.7 Barrier bag, heat seal test.-The test shall be conducted in accordance with the heat seal test specified in Specification MIL-P-116, with a static load of 50 ounces. Determination shall be made for conformance as specified in Specification MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Packaging

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5.1.1 Level A.-The delay or non-delay elements shall be packaged in accordance with Drawing 8796667.

5.2 Packing

5.2.1 Level A.-The delay or non-delay elements shall be packed in accordance with Drawing 8796666.

5.3 Marking.-Marking shall be in accordance with Drawing 8796666 or 8796667, as applicable.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Distribution of data cards shall be as specified by the procuring activity.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

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6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contrator. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35mm microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Unless otherwise specified on the IEL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall

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be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Calculation of standard deviation.-The following formula shall be used to calculate the sample standard deviation:

$$s = \sqrt{\frac{\Sigma(\bar{x} - x)^2}{n - 1}}$$

where:

s is the sample standard deviation.
$\bar{x}$ is the average sample time.
x is each individual time observed.
Σ is the sum of the difference between each observation and the average time squared.
n is the sample size.

6.5 Intended use.-Delay or non-delay elements, M9 are intended for use with Bomb Fuzes, M904E2 and M905.

6.6 Check test for primer deterioration.-The contracting officer shall arrange for the contractor to be reimbursed for the expense of incurring the performance of the check test. The test shall be conducted at Government expense without cost to the contractor assembling the primer into the delay or non-delay element, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness on handling, storage, etc., permitted while the primer lot was under jurisdiction of either contractor.

6.7 Method of manufacture.-Suitability of the igniter composition and delay composition method of manufacture shall be determined from the ability of each composition and of the loaded delay element assembly to meet the requirements specified. After the method of manufacture has been established and placed in production, there shall be no basic change in this method without prior approval of the procuring activity. Records shall be maintained to show the method employed for each lot of loaded and assembled delay elements.

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6.8 Warning.-Both powdered zirconium and the igniter composition are very sensitive to initiation by the discharge of static electricity, as well as by the application of heat or flame. The igniter composition is also very sensitive to friction and impact forces and has been known to ignite when poured. The principle dangers resulting from the ignition of the dry powders are the intense heat developed and the spattering of the molten slag which is formed upon combustion. However, when either zirconium or igniter composition are wet with quantities of water up to 10 or 20 percent by weight, the reaction becomes explosive in nature due to the reaction between zirconium and water to form hydrogen gas. The hazard is also increased when the igniter composition is wet with an inflammable solvent before initiation. The intense heat of the burning composition vaporizes the solvent and a flash fire results. Operators should be provided with suitable protective covering when handling more than 10 grams of either material. Because of the hazards involved, it is recommended that the container size be limited to approximately one pound.

6.9 Test data.-Two copies of the test data for each lot are to be forwarded to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-ND4.

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